arXiv:2407.10953v3 [cs.CL] 15 Dec 2024

Point and Line: Multilingual Mutual Reinforcement Effect Mix Information Extraction Datasets

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Abstract

The Mutual Reinforcement Effect (MRE) represents a promising avenue in information extraction and multitasking research. Nevertheless, its applicability has been constrained due to the exclusive availability of MRE mix datasets in Japanese, thereby limiting comprehensive exploration by the global research community. To address this limitation, we introduce a Multilingual MRE mix dataset (MMM) that encompasses 21 sub-datasets in English, Japanese, and Chinese. In this paper, we also propose a method for dataset translation assisted by Large Language Models (LLMs), which significantly reduces the manual annotation time required for dataset construction by leveraging LLMs to translate the original Japanese datasets. Additionally, we have enriched the dataset by incorporating open-domain Named Entity Recognition (NER) and sentence classification tasks. Utilizing this expanded dataset, we developed a unified input-output framework to train an Open-domain Information Extraction Large Language Model (OIELLM). The OIELLM model demonstrates the capability to effectively process novel MMM datasets, exhibiting significant improvements in performance. The OIELLM model and datasets is open-source in HuggingFace: GitHub Website*

1 Introduction

Information extraction (IE) Sarawagi et al. (2008) is a significant area of research within natural language processing (NLP). This field has evolved to encompass a variety of subtasks, including sentence classification (), text classification (), Named Entity Recognition (NER) (Qu et al., 2023; Nadeau and Sekine, 2007; Lample et al., 2016), sentiment analysis (Tan et al., 2023; Medhat et al., 2014; Rodríguez-Ibánez et al., 2023), relationship extraction (Wadhwa et al., 2023; Mintz et al., 2009; Etzioni et al., 2008), and event extraction (Gao et al., 2023; Xiang and Wang, 2019). Traditionally, these IE subtasks have been segregated into distinct categories for processing. In conventional multi-task IE (Sun et al., 2023; Zhao et al., 2020), datasets from various tasks are typically merged and subsequently fine-tuned using a unified model. This process culminates in the extraction of information from multiple subtasks, each directed by task-specific output heads. While this method effectively leverages the internal knowledge of the model across different IE tasks, it does not address the potential interconnections among the tasks themselves. This omission highlights a gap in understanding how these tasks might benefit from exploring their mutual relationships.

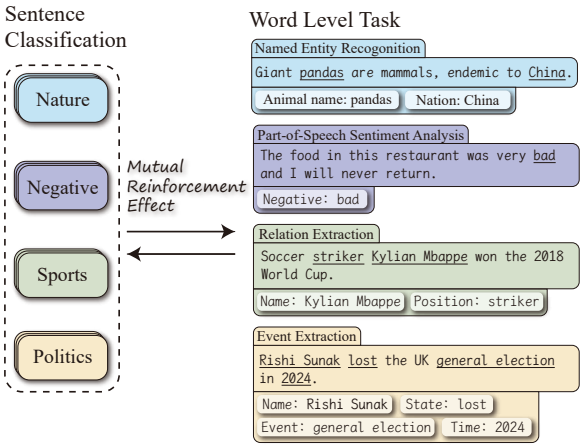

Figure 1: The Mutual Reinforcement Effect between the labels of Word-level labels and text-level label within a same text. **A word-level IE task is a Point, and a text-level IE task is a Line. There is Mutual Reinforcement Effect between the point and the line.**

The Mutual Reinforcement Effect (MRE) Gan et al. (2023b) introduces a novel approach in multitasking IE, emphasizing task interconnections to enhance performance. MRE categorizes IE sub-

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tasks into text-level tasks (e.g., sentence classification, text sentiment analysis) and word-level tasks (e.g., NER). Unlike conventional IE multitasking, which extracts data from various texts, MRE simultaneously performs text-level classification and word-level label-entities pairing within the same text.

MRE categorizes IE tasks into word-level and text-level tasks, analogous to points and lines. Understanding either part helps reinforce the comprehension of the other. Traditionally, IE subtasks have been studied separately, focusing either on points or lines. MRE, however, is the first approach to integrate these two levels, exploring their interdependencies. This not only enhances the performance of IE subtasks but also has implications for future LLM training. When training data is limited, MRE enables dual-level training of LLMs using a single dataset, maximizing its utility and improving model performance.

Figure 1 illustrates MRE in action. The left side depicts sentence classification labels, while the right side shows words with their corresponding labels, representing text-level and word-level tasks, respectively. For example, the sentence 'Giant pandas are mammals, endemic to China.' is labeled 'nature' and contains entity pairs 'Animal Name: pandas' and 'Nation: China.' This highlights how text-level classification and word-level entity recognition reinforce each other, improving accuracy.

Similarly, in sentiment analysis, a text with many positive words likely conveys a positive sentiment. Conversely, a negative-text classification indicates the presence of negative words. This interaction mirrors human text comprehension, where meaning is derived from individual words and synthesized into an overall context (Gan et al., 2023c).

Figure 2 shows the composition of the Multilingual Mutual Reinforcement Effect Mix (MMM) Datasets, which include seven subdatasets per language across three languages. Notably, SCPOS, focused on sentiment classification and part-of-speech tagging, is larger than others and thus not depicted proportionally. SCNM involves sentence classification and NER, while TCREE covers text classification, relation, and event extraction. TCONER leverages an open-domain dataset for text classification and NER.

We translated six MRE mix datasets and expanded the TCONER dataset. To improve LLM performance on IE tasks, we refined the training

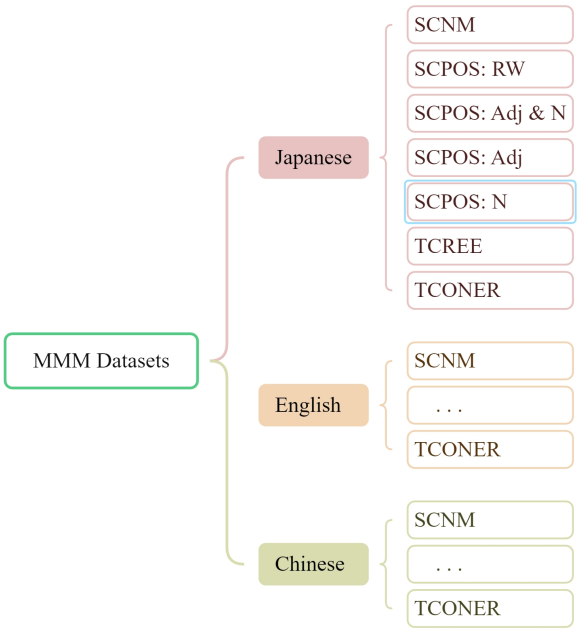

Figure 2: Multilingual Mutual Reinforcement Effect Mix Datasets Names of all sub-datasets. (The image does not represent a percentage of the actual subdataset size.)

process by introducing a streamlined input-output scheme, standardizing task handling, and training the LLM with the MMM dataset. The resulting optimized model, OIELLM, outperformed previous models on multiple datasets, demonstrating the effectiveness of using expanded MRE mix datasets.

Key contributions include:

1. Proposing a framework that uses existing LLMs to translate datasets for underrepresented languages, facilitating multilingual MRE Mix Dataset construction and reducing reliance on manual labor.

2. Expanding the MRE Mix datasets with a newly constructed open-domain NER dataset, enhancing coverage and utility for open-domain applications.

3. Optimizing IE LLM input-output mechanisms using the MMM dataset, surpassing standard models trained on generic data and improving LLM performance in IE tasks.

2 Related Work

Datasets. To begin, the MRE mix dataset primarily originates from the SCNM Gan et al. (2023b) dataset in Japanese, followed by the SCPOS (Gan et al., 2023d) and TCREE Gan et al.

(2023a) datasets. However, the exclusive use of the Japanese language across these datasets poses significant challenges for researchers attempting to further explore the MRE. Moreover, there has been a growing interest in employing LLMs for dataset construction (Tan et al., 2024; Wadhwa et al., 2023; Li et al., 2023; Laskar et al., 2023). Pioneering studies Huang et al. (2023) have demonstrated the efficacy of LLMs in data annotation, where LLM-annotated datasets have outperformed manually annotated counterparts. For instance, LLMs have been utilized to generate datasets for mathematical problems Lin et al. (2024) and to develop dataset labeling frameworks, such as FreeAL (Xiao et al., 2023a), where the data is initially labeled by LLMs and subsequently refined by smaller models before undergoing a final, more accurate labeling by LLMs again.

These methodologies leverage instructional learning and in-context learning to guide LLMs to respond to specific queries and from these responses, extract annotated labels, thereby creating a fully labeled dataset. Distinct from previous efforts, the MMM dataset represents the inaugural initiative to translate datasets from lesser-used languages into more widely spoken languages, such as English and Chinese. Furthermore, the newly developed TCONER dataset addresses a critical gap by providing the first open-domain Named Entity Recognition (NER) dataset within the existing framework of the MRE mix dataset.

LLM in Information Extraction. Since the introduction of Pretrained Language Models (PLMs), sequential-to-sequential (seq2seq) based IE models have gained prominence. These developments range from the initial UIE Lu et al. (2022) to later models such as USM Lou et al. (2023) and Mirror (Zhu et al., 2023). All these models are generative in nature, enabling them to handle multiple word-level IE tasks—such as NER, Relation Extraction, and Event Extraction simultaneously. The primary advantage of these generative IE models is their generalizability; they eliminate the need for task-specific fine-tuning across different tasks. Instead, a single model can address all IE subtasks by standardizing the format of inputs and outputs for various tasks. The model is trained across different IE subtasks using these unified formats, aiming to equip a single model with the capability to manage multiple tasks effectively.

With the advent of LLMs, new approaches to IE have emerged, which can be broadly divided into two categories. The first involves direct interaction with LLMs using prompts in a zero-shot or few-shot manner, where the model outputs the desired entities either through multi-round dialog-style prompts or through single-command-based prompts that extract entities in one go (Wang et al., 2023; Wei et al., 2023). The second approach involves fine-tuning LLMs using specialized datasets (Zhou et al., 2023; Xiao et al., 2023b).

Our research distinguishes itself by focusing more intensively on the MRE. We go beyond merely aggregating existing IE sub-datasets for model training. Instead, we develop specialized MRE-enhanced datasets, through which we not only demonstrate but also apply the efficacy of MRE in enhancing information extraction capabilities.

3 Multilingual Mutual Reinforcement Effect Mix Datasets

In this chapter we will explain how to translate MRE mix datasets in small languages into other languages. And how to construct TCONER datasets. And how you can minimize the use of manual labor with guaranteed quality.

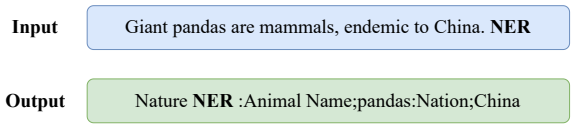

Figure 3: The format of MMM datasets.

3.1 Dataset Translation Framework

First, it is essential to understand the format of the Multilingual Mutual Reinforcement Effect Mix (MMM) dataset. As depicted in Figure 3, the MMM dataset comprises inputs and outputs. The input section, highlighted in blue, includes both text and a task instruction word, such as "NER." In the output section, shown in green, the initial output is a text-level classification label, followed by the task instruction word "NER." The labeling follows the start and end symbols (i.e., ":", ";") used in the original MRE mixed dataset. This format allows for consistent generation of label-entity pairs regardless of quantity (e.g., ":label1;entities1:label2;entities2..."). Thus, the task instruction word guides the model in producing various word-level extracted information alongside the text-level classification label.

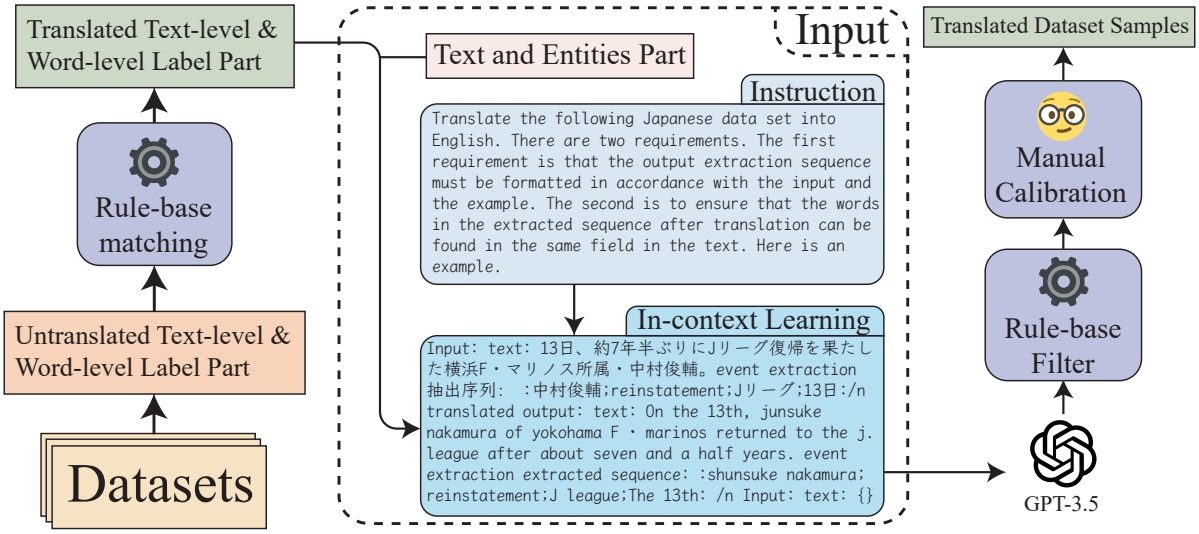

Figure 4: The overview of dataset translation framework.

Figure 4 presents a flowchart of the entire dataset translation framework. The process begins on the leftmost side, where six sub-datasets are initially processed using a rule-based matching method, according to their classifications. The labels at both text and word levels are systematically translated into English and Chinese. Given the consistent labeling across datasets, this translation can proceed directly based on predefined rules. For instance, the Japanese label "ポジティブ" is directly translated as "positive." Employing a rule-based approach for label translation is not only quick and precise but also simplifies the subsequent translation of text and entities. Furthermore, these translated labels are input into a LLM along with the untranslated text and entities, serving an auxiliary role in the translation process.

The process involves two main inputs to the LLM, GPT-3.5-Turbo Ouyang et al. (2022): the part with translated labels and the part with untranslated text and entities. We employ both instruction-based and in-context learning (ICL) methodologies for this translation task. As depicted in the central portion of Figure 4, the selection of the instruction template was refined through multiple iterations. Initially, a simple instruction such as "Translate the following Japanese dataset into English." failed to produce satisfactory translations. Consequently, we introduced several constraints to enhance the output quality. These include stipulating that the model's output format must align with the example provided below, with a critical requirement being the accurate translation of entities, ensuring they correspond directly to terms found in the original Japanese text. Additional constraints were applied specifically for Japanese-to-Chinese translations, such as informing the model that labels have been pre-translated and only text and entities require translation. We also instructed the model to ensure comprehensive translation into Chinese. Furthermore, a one-shot example of ICL was provided to demonstrate the desired outcome, guiding the model to generate translations strictly adhering to the specified format.

Finally, we obtained the translated dataset. However, due to the inherent unpredictability of LLM outputs, it is not always guaranteed that the outputs will conform to the expected format, even when the inputs are consistent. To address this, we implemented a dual-component rule-based filtering mechanism. The first component involves removing samples containing any residual Japanese characters from the translated data. The second component entails verifying whether the translated entities exactly match words in the text. Samples that do not meet this criterion are excluded. Additionally, this step assesses whether the pairings of labels and entities adhere to the formatting standards of the MMM dataset.

Despite the substantial reduction in dataset size resulting from the first two steps—translation and filtering—the remaining data exhibit exceptionally high translation quality. The final dataset undergoes a manual review and correction process, which ensures maximum accuracy while minimizing the reliance on manual labor. This approach outlines our tailored dataset translation framework, designed to accommodate the specific characteris-

tics of the MMM dataset. With minimal modifications, this framework can be adapted for translating datasets for other tasks, effectively addressing the scarcity of datasets in lesser-used languages.

3.2 Construction of TCONER

In the original MRE mix datasets, relation and event extraction tasks are open-domain, implying that the labels are not predefined. However, the label set is limited to only a dozen options. Given this context, we constructed a new dataset, termed TCONER, based on an open-domain Named Entity Recognition (NER) dataset[†] (Zhou et al., 2023). The labels at the text level in the TCONER dataset are also open-domain. To annotate this dataset, we initially employed the GPT-3.5-Turbo model to assign open-domain text-level labels. Subsequent manual verification and annotation were conducted to ensure accuracy and consistency, resulting in the finalized TCONER dataset. Similarly, we translated the constructed English TCONER dataset using the dataset translation framework. The TCONER dataset was translated into Japanese and Chinese.

3.3 Results of Datasets Construction

Table 1 presents the statistics of the final translation results. Due to the high costs associated with the use of a premium API, we limited our study to 10,000 samples from each of three sub-datasets within SCPOS and the TCONER dataset, which contains 180,000 entries. These 10,000 samples, retained post-translation, proved to be an ample test set. It was observed that there was a greater data loss when translating into Chinese compared to English. This discrepancy may be attributed to the training data predominance of English in OpenAI's GPT-3.5-Turbo model, resulting in superior performance in English-related tasks. For instance, in the SCNM and TCREE datasets, the Japanese to English translation accuracy exceeded 80%. Conversely, the translation results from English to Chinese in the TCONER dataset were markedly better than those from English to Japanese. This further confirms that GPT-3.5-Turbo exhibits enhanced effectiveness with major languages compared to lesser-used ones.

[†] https://huggingface.co/datasets/Universal-NER/Pile-NER-type?row=0

Dataset	Japanese	English	Chinese
SCNM	**5343**	4449	3177
SCPOS: RW	**2000**	1312	1406
SCPOS: Adj & N	**187528**	4801	3937
SCPOS: Adj	**187528**	9132	7413
SCPOS: N	**187528**	5027	3920
TCREE	**2000**	1910	1491
TCONER	6791	**45888**	9047

Table 1: Statistical results of the translated MMM dataset. (Due to resource constraints, we extracted only 10,000 samples as translation objects from each of the three SCPOS sub-datasets and the TCONER dataset.)

4 Open-domain Information Extraction Large Language Model

In this chapter, we outline methodologies to enhance the performance of existing models and techniques for processing MRE mix datasets, aiming to surpass previous benchmarks. Before delving into the specifics of the Open-domain Information Extraction Large Language Model (OIELLM), it is imperative to justify the necessity for a distinct model tailored to MMM datasets.

Firstly, MRE mix datasets differ significantly from traditional IE tasks as they require simultaneous output of text-level labels and word-level label-entity pairs. Consequently, standard sequence labeling models are inadequate for handling these demands directly. Furthermore, existing generative IE models and methodologies have solely focused on producing word-level label-entities, neglecting text-level labels altogether.

The primary objective of MRE mix datasets is to investigate the interplay between text-level and word-level annotations. By leveraging this synergistic relationship, we aim to concurrently enhance the performance of both tasks. This model improves textual understanding by learning both tasks in tandem. Additionally, the MRE framework can contribute to model interpretability, drawing inspiration from cognitive processes that mimic human reasoning.

In summary, this study proposes the development of a novel model specifically designed for processing the MMM dataset. Furthermore, we aim to experimentally investigate whether the MRE positively influences various IE subtasks when using LLMs. Traditionally, IE tasks have been approached through the use of QA dialogues for extraction. However, this research adopts a distinct methodology. This departure is motivated by earlier foundational studies in generic generative IE

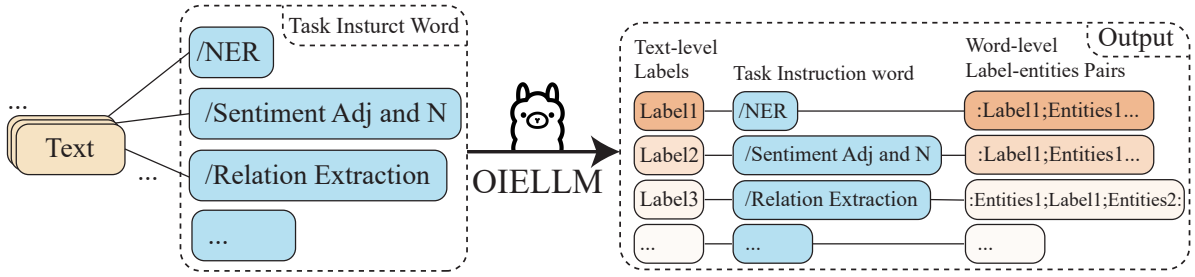

Figure 5: The input and output of Open-domain Information Extraction Large Language Model (OIELLM).

with PLMs, where dialogue models were not utilized. Instead, these studies implemented a generic framework. Accordingly, we too will employ a modified input and output scheme tailored for the MMM dataset, diverging from the conventional dialogue-based approaches.

Figure 5 illustrates the input and output formats of our enhanced OIELLM. The fundamental unit of analysis in both input and output is words, reflecting our understanding of the tokenization principle utilized by LLMs, which typically focuses on words or phrases. By omitting the dialog prompt, we do not compromise the LLM's comprehension of the task. This adjustment not only reduces the input-output length but also simplifies the LLM's processing, thereby enhancing operational speed.

Each text processed is prefixed with task-specific instruction words, which define the task type and guide the model's subsequent output generation. In our format, all task instruction words in the input are introduced by a special symbol "/", which serves to delineate the task words from the main text. This separation is crucial for distinguishing between text-level labels and word-level label-entity pairs in the output.

The combined text and task instruction words are then fed into the OIELLM, with the output comprising both text-level labels and word-level label-entity pairs. Our labeling convention adheres to the format used in the previous MRE mix datasets, utilizing ":" and ";" to ensure consistency and clarity.

In summary, by standardizing the input and output structures and clearly defining task instruction words, our modified OIELLM effectively processes all sub-datasets within the MMM framework.

5 Experiment

In this chapter, we detail specific experimental procedures, including dataset sizes for the MMM dataset and methodologies for training the OIELLM model, along with the evaluation techniques used.

5.1 Details of OIELLM Training

We began by selecting baselines: USA-7B (IL + ICL)[‡] and GIELLM-13B-jp[§], previously utilized for processing the MRE mixed datasets, served as comparative models. For the foundational architecture of OIELLM, we chose the latest Instruct and Base version of LLaMA3-8B[¶]. Since LLaMA3 does not offer a 13B version, we incorporated the LLaMA2-13B Touvron et al. (2023) model as well.

We attempted to evaluate the MMM dataset using the GPT-3.5-Turbo model; however, this model failed to produce the expected information and was unable to maintain a consistent format, despite being provided with an adequate number of few-shot examples for training. The resulting F1-score was near zero. Consequently, we decided not to select the GPT-3.5-Turbo model for further testing in our study.

OIELLM was fine-tuned using full parameters based on these three models. Training was conducted at BF16 precision, while inference was performed at FP16. The training spanned 3 epochs with a learning rate of 1e-5, utilizing computational resources including three A800 80GB and three RTX 6000 Ada 48GB GPUs, with training durations ranging from 12 to 20 hours. For the training and test sets, Comprehensive statistics on the training and test sets are available in Table 3, 4.

6 Statistical Results of Train and Test Dataset in OIELLM

As shown in Tables 3 and 4, the statistics for the complete training and test sets of the MMM dataset.

[‡] https://huggingface.co/ganchengguang/USA-7B-instruction-incontext-learning

[§] https://huggingface.co/ganchengguang/GIELLM-13B-jpllm

[¶] https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct

Japanese	SCNM			SCPOS: RW			SCPOS: Adj & N		
Model	TL	WL	ALL	TL	WL	ALL	TL	WL	ALL
GPT-3.5-Turbo	42.07	7.54	1.97	57.20	0	0	28.97	5.97	0
USA-7B	-	-	-	53.27	40.80	7.67	91.33	**81.68**	**9.63**
GIELLM-13B-jp	85.47	84.46	54.2	86.01	**66.61**	**17.39**	93.23	47.35	0.20
OIELLM-8B	84.73	88.53	61.93	86.50	54.76	12.40	89.13	14.88	0.40
OIELLM-8B*	87.30	**89.28**	**64.00**	88.20	53.79	12.30	89.63	15.84	0.73
OIELLM-13B	**89.00**	86.33	57.70	**94.60**	52.36	11.90	**95.20**	11.94	0.20
Japanese	SCPOS: Adj			SCPOS: N			TCREE		
Model	TL	WL	ALL	TL	WL	ALL	TL	WL	ALL
GPT-3.5-Turbo	65.50	0.31	0.87	39.60	6.79	0	57.20	0	0
USA-7B	91.43	45.51	51.77	92.03	**81.30**	**9.73**	-	-	-
GIELLM-13B-jp	93.67	45.06	**55.67**	92.83	46.42	0.33	**97.47**	**79.01**	77.89
OIELLM-8B	87.13	74.96	53.07	87.77	22.92	0.50	95.07	74.92	83.69
OIELLM-8B*	89.93	**75.33**	54.93	90.63	23.69	0.63	96.98	74.42	**84.19**
OIELLM-13B	**94.00**	60.69	42.50	**94.70**	18.07	0.60	97.08	73.82	84.19

English	SCNM			SCPOS: RW			SCPOS: Adj & N		
Model	TL	WL	ALL	TL	WL	ALL	TL	WL	ALL
GPT-3.5-Turbo	53.50	0.04	0	14.78	2.11	0.12	68.63	13.62	0.33
OIELLM-8B	82.30	81.36	52.53	72.17	49.60	11.82	76.57	18.00	1.67
OIELLM-8B*	**85.43**	**82.38**	**55.43**	**74.75**	**49.93**	**12.81**	79.77	**19.28**	2.27
OIELLM-13B	84.80	80.68	50.60	95.07	46.64	12.19	**94.30**	18.59	**3.20**
English	SCPOS: Adj			SCPOS: N			TCREE		
Model	TL	WL	ALL	TL	WL	ALL	TL	WL	ALL
GPT-3.5-Turbo	6.97	0.26	0.03	0.53	0.08	0	12.87	0	0
OIELLM-8B	75.47	51.85	32.33	76.10	28.67	1.27	80.87	21.77	33.67
OIELLM-8B*	76.60	**51.95**	33.17	78.67	27.45	**0.73**	80.23	**25.90**	22.37
OIELLM-13B	**94.40**	50.56	**38.40**	**95.30**	**28.36**	0.60	**89.90**	23.50	**22.60**

Chinese	SCNM			SCPOS: RW			SCPOS: Adj & N		
Model	TL	WL	ALL	TL	WL	ALL	TL	WL	ALL
GPT-3.5-Turbo	41.63	9.57	2.30	50.77	2.08	0.78	59.33	7.18	0.40
OIELLM-8B	84.90	**71.90**	46.40	89.29	45.75	9.93	92.33	8.75	0.33
OIELLM-8B*	86.33	69.97	**46.77**	92.27	**46.20**	**10.60**	94.50	**8.46**	0.40
OIELLM-13B	**87.70**	68.12	41.60	**95.03**	43.32	8.72	**94.90**	8.42	**0.50**
Chinese	SCPOS: Adj			SCPOS: N			TCREE		
Model	TL	WL	ALL	TL	WL	ALL	TL	WL	ALL
GPT-3.5-Turbo	56.27	0.19	0.07	53.07	3.11	0.53	59.33	7.18	0.40
OIELLM-8B	93.73	60.96	53.00	92.63	28.32	0.63	91.73	58.12	56.41
OIELLM-8B*	95.80	64.51	**57.63**	94.97	**28.91**	**1.30**	95.06	**59.54**	**58.83**
OIELLM-13B	**96.00**	60.68	54.90	**95.20**	27.77	1.00	**95.26**	56.91	56.00

TCONER	English			Japanese			Chinese		
Model	TL	WL	ALL	TL	WL	ALL	TL	WL	ALL
GPT-3.5-Turbo	23.87	4.78	0	23.87	2.24	0.17	29.47	2.97	**0.57**
OIELLM-8B	24.80	21.12	0.20	27.70	13.83	**0.20**	33.73	**18.87**	0
OIELLM-8B*	37.13	**23.05**	**0.30**	41.40	**14.24**	0.17	**48.27**	18.06	0.17
OIELLM-13B	**40.30**	19.23	**0.30**	**43.40**	13.02	0	47.70	15.72	0.30

Table 2: The F1 score of MMM datasets. TL: Text-Level. WL: Word-level. ALL: TL and WL are correct simultaneously.

The MMM dataset was segmented into 21 sub-datasets. Training set sizes were assigned based on the sizes of these sub-datasets, categorized into three groups: 500, 1000, and 2000 samples. Samples beyond these numbers were allocated to the test sets.

Dataset	Japanese	English	Chinese
SCNM	1000	1000	1000
SCPOS: RW	1000	500	500
SCPOS: Adj & N	1000	1000	1000
SCPOS: Adj	1000	1000	1000
SCPOS: N	1000	1000	1000
TCREE	1000	500	500
TCONER	2000	2000	2000

Table 3: Statistical results of train sets of OIELLM.

Dataset	Japanese	English	Chinese
SCNM	4343	3449	2177
SCPOS: RW	1000	812	906
SCPOS: Adj & N	186528	3801	2937
SCPOS: Adj	186528	8132	6413
SCPOS: N	186528	4027	2920
TCREE	1000	1410	991
TCONER	4791	43888	7047

Table 4: Statistical results of test sets.

6.1 Evaluation

We employed the F1 score as our primary metric for evaluation. Initially, the model's output was bifurcated into two segments based on the task-specific instruct word: the Text-level Label and the Label-entities pairs. Subsequently, Label-entities pairs were delimited using start-end symbols (i.e., ":", ";"). Each Label-entity pair was treated as an individual element within the set. The F1 score was segmented into three categories: Text-level (TL), Word-level (WL), and ALL. These represent the F1 scores at respective levels and the aggregate F1 score when both levels are accurately predicted in an output. The formulas for calculating the F1 score from the set of segmented label-entities pairs are shown in Equations (1) - (3) below. See Appendix A for specific pseudo-code for the entire evaluation.

$$\text{Precision} = \frac{|Real \cap Generated|}{|Generated|} \tag{1}$$

$$\text{Recall} = \frac{|Real \cap Generated|}{|Real|} \tag{2}$$

$$F_1 \text{ Score} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{3}$$

7 Results

Table 2 presents the experimental results of three OIELLM models trained on 21 MMM sub-datasets. Notably, the model designated with an asterisk, OIELLM-8B, was trained using the LLaMA3-8B-Instruct framework, whereas the remaining models were based on the LLaMA3-8B-Base framework. These results demonstrate the enhanced performance of OIELLM in handling Japanese data after incorporating multilingual capabilities. Impressively, OIELLM's performance surpassed that of GIELLM-13B-jp on half of the datasets, despite GIELLM-13B-jp being a model specifically tailored for Japanese. This observation supports the hypothesis that integrating multilingualism and multitasking can more effectively leverage the knowledge embedded in the pre-training of multilingual LLMs.

However, OIELLM's performance on the TCONER task was suboptimal, which we attribute to insufficient training data. Given that open-domain tasks require extensive and diverse datasets compared to domain-specific tasks, the limited data may have hindered the model's performance. This area will be a focus of our future research, aiming to understand and improve the data dependencies of OIELLM in open-domain contexts. Due to the high cost of accessing GPT-4/o, we conducted experiments on MMM datasets using GPT-3.5-Turbo only. Despite tailoring prompt templates to align with each dataset's specific features, GPT-3.5-Turbo demonstrated subpar performance. This can be attributed to the model's limited suitability for tasks requiring the simultaneous extraction of multiple label-entity pairs. It frequently generates redundant fields and extraneous characters, resulting in a lower F1 score. These limitations underscore the necessity of developing an OIELLM specifically trained for this task.

8 Conclusion and Future Work

In this study, we introduce a framework that utilizes LLMs to translate datasets, thereby removing language barriers for research in less-represented languages. To address the deficiency of open-domain IE tasks in the MRE mix dataset, we constructed the TCONER dataset. Additionally, we trained the OIELLM model using the newly created MMM dataset.

Future work will focus on employing the MMM dataset to further explore the Mutual Reinforcement Effect. We will also continue to enhance the performance of the OIELLM model in open-domain information extraction tasks.

9 Limitations

Due to resource constraints, we were unable to employ the higher-performing GPT-4-Turbo OpenAI

(2023) model as the base for our dataset translation framework. Consequently, this model was also not utilized during the testing phase on the dataset. In future work, we aim to leverage a more advanced model, such as the GPT-4-Turbo, to evaluate the MMM dataset, provided that the necessary resources become available.

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A Evaluation Pseud Code

Algorithm 1 Parse Text Label and Entity Pairs

```
procedure PARSE_OUTPUT(output, instruct_word, is_tcree)
    Input: output (String), instruct_word (String), is_tcree (Boolean)
    Output: text_label (String), entity_pairs (Set of Tuples)

    instruct_word ← instruct_word
    if instruct_word ∉ output then
        return ("", {})
    end if
    text_label, entity_pairs ← output.split(instruct_word, 1)
    text_label ← text_label.strip()
    if is_tcree then
        entity_pairs ← [entity_pairs.strip()]
    else
        entity_pairs ← [pair.strip() for pair in entity_pairs.split(" : ") if pair]
    end if
    entity_pairs ← [tuple(pair.split(";")) for pair in entity_pairs]
    return (text_label, set(entity_pairs))
end procedure
```

B Case Study of Input and Output Format with OIELLM in MRE mix datasets

Input **Output**

SCNM: Sentence Classification and Named Entity Recognition Mix Dataset

2018年からクリシューマECで主力メンバーとして活動したが、2020年1月18日、日本のレノファ山口FCへの加入が発表された。/固有表現抽出

文芸/固有表現抽出/:その他の組織名;クリシューマEC:地名;日本:その他の組織名;レノファ山口FC

SCPOS: RW: Sentiment Text Classification and Part-of-Speech: Related Words Mix Dataset

純正の商品なのでとっても使いやすいです。ふたを裏側にしても磁石でつくので邪魔になりません。/感情分析名詞

積極/感情分析名詞/:ポジティブ;純正:ポジティブ;やすい:ポジティブ;邪魔になりません

SCPOS: Adj & N: Sentiment Text Classification and Part-of-Speech: Adjective & Noun Mix Dataset

このバンドを聴いたのは本作が初。印象はというと、オールドロックながらも随所に彼等流の新開拓を施しており、古臭みなく聴ける内容となっている。まだ数回しか聴いてないので断定した言い方はできないが、ハッキリ言ってカッコいい。ロックバンドでも数本の指に入るくらい気に入りましたね。/感情分析形容詞名詞"

積極/感情分析形容詞名詞/:ポジティブ;いい:ポジティブ;カッコいい:中立;たね:ネガティブ;らい:ネガティブ;ドロ:中立;印:中立;回し:中立;開拓:中立;言い方:中立;新:中立;断定:中立;内

SCPOS: Adj: Sentiment Text Classification and Part-of-Speech: Adjective Mix Dataset

個人的にKID Aより好きかもしれない。たぶん一番好き。何よりもわかりやすい。まさに姉妹アルバム。オリジナリティ溢れる作品。かなりロマンティックです。/感情分析形容詞

積極/感情分析形容詞/:ポジティブ;わかりやすい

SCPOS: N: Sentiment Text Classification and Part-of-Speech: Noun Mix Dataset

グリム レジェンド2。映像は綺麗で、ストーリーも良く出来ています。ミニゲームも普通ですね。三作目も作っているそうなので、期待して待っています。/感情分析名詞

積極/感情分析名詞/:中立;ゲーム:ネガティブ;スト:ポジティブ;期待:ポジティブ;出来:ポジティブ;通:中立;普通:中立;目:ポジティブ;良:ポジティブ;綺麗

TCREE: Text Classification and Relation & Event Extraction Mix Dataset

2012年に公開を控える新作映画の中で、MOVIE ENTER編集部オススメの作品を紹介する「2012年先取り映画」。今回“ドラマのサイトー”が紹介する作品は『わが母の記』です。主演の役所広司が出演しています。/関係抽出

映画/関係抽出/:役所広司;主演;わが母の記:

TCONER: Open-domain Text Classification and Named Entity Recognition Mix Dataset

ルネサンス・ソサエティ・オブ・アメリカについて\n\nルネサンス・ソサエティ・オブ・アメリカは1954年に設立され、1300年から1650年の時代の研究を推進しています。RSAは北アメリカや世界中からさまざまな専門分野の学者を集めています。RSAには、大学やカレッジの教授、講師、大学院生、博物館、図書館、他の文化機関のメンバーが5,000人以上おり、独立した学者やルネサンス研究に興味を持つ他の多くの人々もいます。/固有表現抽出

教育/固有表現抽出/:略語;RSA:日付;1954:組織;ルネサンス・ソサエティ・オブ・アメリカ:組織;大学:組織;カレッジ:組織;博物館:組織;図書館:組織;文化機関:職業;教授:職業;講師:職業;大学院生:職業;独立した学者:場所;北アメリカ:場所;世界:主題;ルネサンス研究:数;5,000

Figure 6: The input and output format example with OIELLM in Japanese MRE mix datasets.

Input Output

SCNM: Sentence Classification and Named Entity Recognition Mix Dataset

Since 1989, Sanrio has been using \"Minna no Taabo\" as a character, and in the 1990s, they used Miho Kanno, Mariru Watanabe, Hideyuki Yakou, who appeared in the Hokkaido-based TV drama \"Kita no Kuni Kara,\" and Yoshiji Masuda, who is from Hokkaido, as CM characters./NER

Literature/NER/:Company;Sanrio:Product Name;Minna no Taabo:Person;Miho Kanno:Person;Mariru Watanabe:Location;Hokkaido:Product Name;Kita no Kuni Kara:Person;Hideyuki Yakou:Location;Hokkaido:Person;Yoshiji Masuda

SCPOS: RW: Sentiment Text Classification and Part-of-Speech: Related Words Mix Dataset

A variety of unique numbers are lined up, and it's never boring to listen to. The diversity is wonderful. I think it will remain for future generations./Sentiment related word

positive/Sentiment related word/:neutral;unique:positive;boring:positive;wonderful

SCPOS: Adj & N: Sentiment Text Classification and Part-of-Speech: Adjective & Noun Mix Dataset

Sample dataset in English:\n\ntext: The wolf, who is usually a bad guy, ... The end is cute and heartwarming, and it's a wonderful story. The Japanese version is also wonderful./Sentiment Adj and N

positive/Sentiment Adj and N/:neutral;story:neutral;wolf:positive;wonderful:negative;bad:negative;bad guy

SCPOS: Adj: Sentiment Text Classification and Part-of-Speech: Adjective Mix Dataset

I felt that all the songs had a slow tempo and the melody was hard to grasp. It seems that there were also some songs used as theme songs, but they were not so great and I did not think they were good. I wish there were more understandable melodies. Perhaps, musical preferences vary by individual? I feel like I wasted a little bit of money purchasing it./Sentiment Adj and N

negative/Sentiment Adj and N/:negative;not so great:positive;good:positive;understandable

SCPOS: N: Sentiment Text Classification and Part-of-Speech: Noun Mix Dataset

It contains my favorite songs, and I bought it because it was cheap. It took so long for it to arrive that I thought it would never come, but there is no problem at all with the content. However, it's a minus one because it took so long./Sentiment N

positive/Sentiment N/:neutral;favorite:positive;cheap:negative;problem

TCREE: Text Classification and Relation & Event Extraction Mix Dataset

The top-selling digital camera from October 11th to 16th was Canon's \"IXY 600F\". /relation extraction

IT/relation extraction/:Canon;Product;IXY 600F:

TCONER: Open-domain Text Classification and Named Entity Recognition Mix Dataset

Drama-documentary exploring the betrayals between the Vikings, Anglo-Saxons and Normans. BBC Two/NER

Entertainment/NER/:organization;BBC Two:group;Vikings:group;Anglo-Saxons:group;Normans

Figure 7: The input and output format example with OIELLM in English MRE mix datasets.

Input Output

SCNM: Sentence Classification and Named Entity Recognition Mix Dataset

几乎在同一时间，村田真等人也在开发与XML Schema不同的新的模式语言RELAX。/实体命名识别

技术/实体命名识别/:人名;村田真:产品名;XML Schema:产品名;RELAX

SCPOS: RW: Sentiment Text Classification and Part-of-Speech: Related Words Mix Dataset

科林·费尔斯赢得了最佳男主角奖，然而此次获奖是因为克里斯蒂安·贝尔是配角，如果他是主角，那么这部电影可能就是毫无存在感的，这部电影根本不值得一提，是部烂片。/感情分析关联单词

消极/感情分析关联单词/:消极;烂片"

SCPOS: Adj & N: Sentiment Text Classification and Part-of-Speech: Adjective & Noun Mix Dataset

虽然不用说内容，但能在美丽的画面和声音中观看，感到很开心。/感情分析形容词名词

积极/感情分析形容词名词/:积极;开心:中立;观看:中立;内容:积极;美丽

SCPOS: Adj: Sentiment Text Classification and Part-of-Speech: Adjective Mix Dataset

虽然是机械式的，但时间被准确地标记，分钟的显示在外面，所以可以快速确认分钟，设计简单轻巧实惠，很好。/感情分析形容词名词

积极/感情分析形容词名词/:积极;好

SCPOS: N: Sentiment Text Classification and Part-of-Speech: Noun Mix Dataset

对于朱莉亚罗伯茨的粉丝来说，这部电影简直让人无法抗拒，休·格兰特也很棒，无论看多少遍都很享受。/感情分析名词

积极/感情分析名词/:积极;棒:积极;粉丝:积极;享受

TCREE: Text Classification and Relation & Event Extraction Mix Dataset

11日深夜，日本电视台的“Going! Sports＆News”节目中，上次参加泳泳太平洋锦标赛时展现出绝对实力，在100米蛙泳和200米蛙泳项目中获得冠军的北岛康介进行了现场直播。/事件抽取

运动/事件抽取/:北岛康介;出演;日本电视台;11日:

TCONER: Open-domain Text Classification and Named Entity Recognition Mix Dataset

预制结构，如所谓的\"移动\"住宅或模块化住宅等，通常通过放置在混凝土或砖块基座上进行安装。通常，这些结构的外围下沿是不受支撑的。这些边缘与地面之间的间隙可能被金属或合成材料制成的非承重裙子覆盖，仅仅是为了提供更加美观的外观。经过很长一段时间，这些建筑的外围边缘可能会翘曲、下垂或变形。裙子对阻止这种形式的恶化并不具备结构上的足够性。/实体命名识别

建筑/实体命名识别/:位置;地面:状态;恶化:结构;模块化住宅:结构;裙子:结构;外围边缘:材料;混凝土:材料;砖块:材料;金属:材料;合成材料

Figure 8: The input and output format example with OIELLM in Chinese MRE mix datasets.